Quantum Optimization for Insurance Fraud Resource Allocation

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From premium pricing to resource alocation

- **The Initial Vision:**
  - Pricing Optimization
  - Optimization for insurance premiums using a QAOA-based approach.

- **Qubit Explosion:**
  - QAOA requires binary decision variables (0 or 1)
  - Total Qubits for pricing problems= (Number of Customers) × (Price Levels to test).

One Hot encoding

- **Interaction Terms:**
  - Representing the impact of interaction in the premiums felt mathematically forced

From premium pricing to resource alocation

- **Optimizing Under Hard Constraints (k):**
  - While automated systems flag thousands of suspicious claims, insurers have a finite "investigative budget" allowing for exactly k manual reviews per cycle

- **From Individual Risk to Network Synergy:**
  - Not just the riskiest individuals, but to select the optimal portfolio of cases taking into account the correlations between the entities

Which contracts do AXA want to control?

Problem modelization QUBO

$$Q(x) = -\sum_{i=1}^{N} v_i x_i - \sum_{i=1}^{N} \sum_{j=i+1}^{N} \omega_{ij} x_i x_j + \lambda \left(\sum_{i=1}^{N} x_i - k\right)^2$$

Problem modelization QUBO

$$Q(x) = -\sum_{i=1}^{N} v_i x_i - \sum_{i=1}^{N} \sum_{j=i+1}^{N} \omega_{ij} x_i x_j + \lambda\left(\sum_{i=1}^{N} x_i - k\right)^2$$

Avoided loss

x_i : case i investigation status (decision variable)

v_i : case i expected gain

Problem modelization QUBO

$$Q(x) = -\sum_{i=1}^{N} v_i x_i - \sum_{i=1}^{N} \sum_{j=i+1}^{N} \omega_{ij} x_i x_j + \lambda\left(\sum_{i=1}^{N} x_i - k\right)^2$$

Interaction term

x_i : case i investigation status (decision variables)

w_{ij} : correlation between case i and case j

Problem modelization QUBO

$$Q(x) = -\sum_{i=1}^{N} v_i x_i - \sum_{i=1}^{N} \sum_{j=i+1}^{N} \omega_{ij} x_i x_j + \lambda\left(\sum_{i=1}^{N} x_i - k\right)^2$$

Hard constraints

x_i : case i investigation status (decision variables)

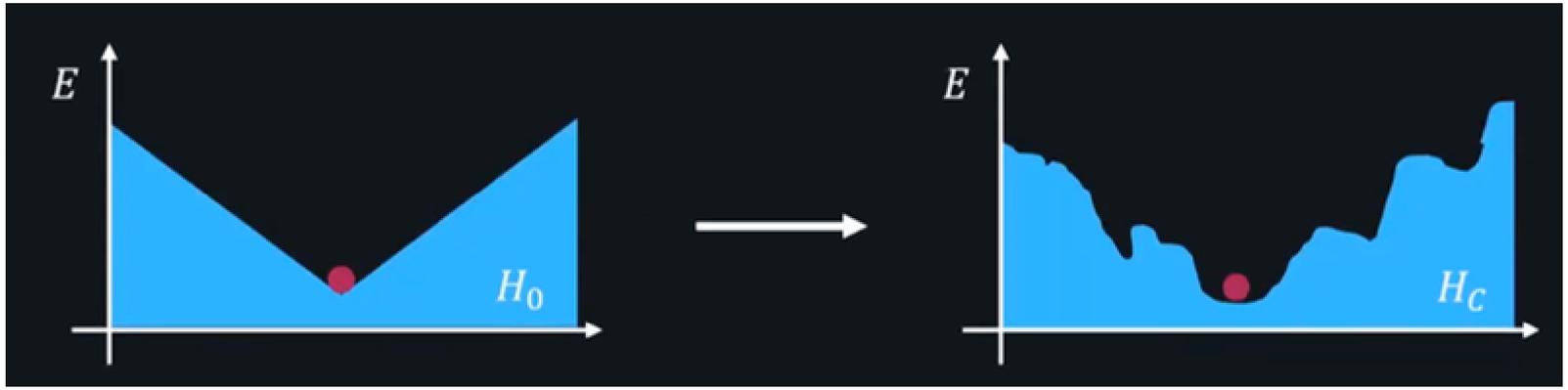
k: max number of investigations

Our dataset

case_id	claim_type	region_id	claim_amour	policy_age_d	prior_claims	complexity	review_minu	broker_id	garage_id	device_id	entity_risk_s	p_fraud	ring_id_laten	severity_L	review_cost	v_i
0	auto	1	3560.23065	3618	0	0.84075578	49	9	3	17	0.18749119	0.05206257	0	1068.06919	49	-18.416464
1	health	0	5940.48244	3255	2	1.45710061	86	0	9	13	0.10655822	0.09844237	0	1188.09649	86	-21.672533
2	auto	3	4609.22976	2689	0	0.57143584	40	7	11	9	-0.2015748	0.04296597	0	1382.76893	40	-7.3233968
3	auto	0	3298.63634	3034	0	1.5115055	72	0	11	13	-0.02122	0.07126588	2	989.590903	72	-33.211762
4	auto	5	3327.93212	3410	0	1.03170472	56	0	10	15	-0.0684511	0.06893724	1	998.379637	56	-18.145956
5	home	2	4912.43948	733	1	1.01072102	63	4	1	22	-0.0386044	0.06398093	0	1964.97579	63	6.1465391
6	auto	5	3040.50773	486	2	0.75642009	62	1	11	33	-0.2950024	0.10518821	2	912.15232	62	-9.2287797
7	auto	4	14539.3017	1532	2	0.4877953	53	6	9	33	-0.0446026	0.15503991	0	4361.7905	53	318.938378
8	auto	0	2438.17208	918	0	0.94306693	53	7	9	24	0.02204169	0.03798778	0	731.451623	53	-37.717577
9	health	4	2069.95115	2203	0	1.39335126	68	1	3	27	0.12413644	0.0533102	3	413.99023	68	-55.861555
10	home	1	18072.5252	3501	0	0.87111287	50	0	3	2	0.22813705	0.16728264	1	7229.01006	50	615.108336
11	health	3	2234.8334	608	0	0.51878647	38	1	6	22	-0.0257749	0.04898817	3	446.966681	38	-25.957156
12	home	3	6589.66371	3023	0	1.00002954	55	3	2	5	0.13953882	0.10112713	2	2635.86549	55	91.6066269
13	auto	3	4501.72351	2087	0	0.61065598	41	0	10	4	0.03448385	0.05143746	0	1350.51705	41	-2.7930546
14	health	0	1127.44243	3056	4	0.89774008	83	8	8	5	0.20353037	0.09422156	0	225.488485	83	-71.314768
15	auto	1	8178.96248	1879	0	0.91710241	52	6	7	9	-0.2112316	0.05599407	0	2453.68874	52	23.5656162

Creation of the **interaction** terms and potential **gains**

QAOA and its great use of physics



Instead of continuous slow evolution, QAOA approximates it by alternating operators:

$$|\psi(\gamma,\beta)\rangle = U_M(\beta_p)U_C(\gamma_p)\cdots U_M(\beta_1)U_C(\gamma_1)|\psi_0\rangle$$

Where:

$$U_C(\gamma) = e^{-i\gamma H_C}$$

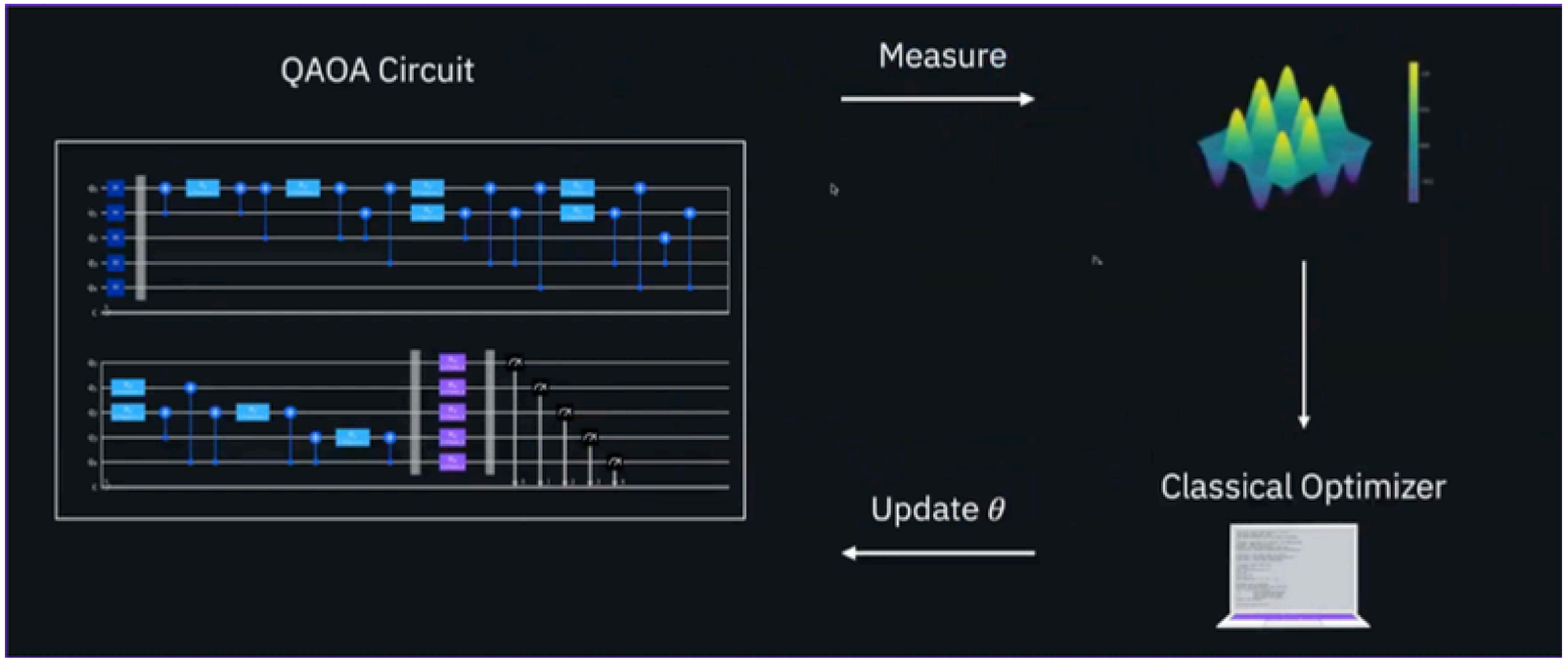
It shifts phases depending on energy
Low-energy states accumulate different phases than high-energy states.

$$U_M(\beta) = e^{-i\beta H_M}$$

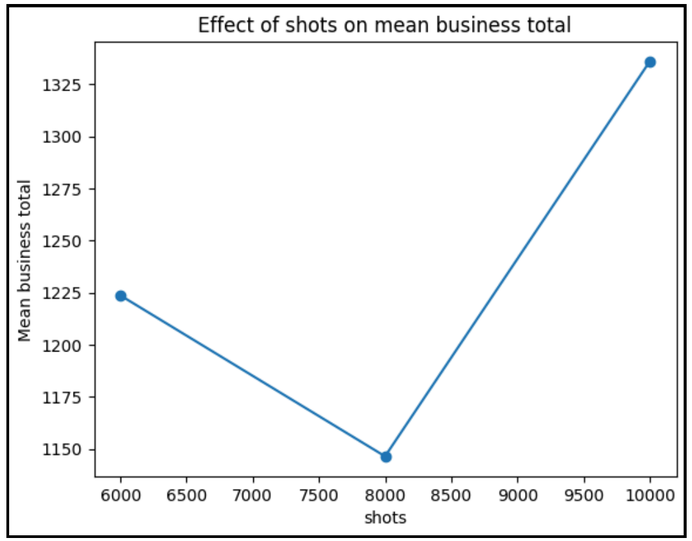
Causes interference between states

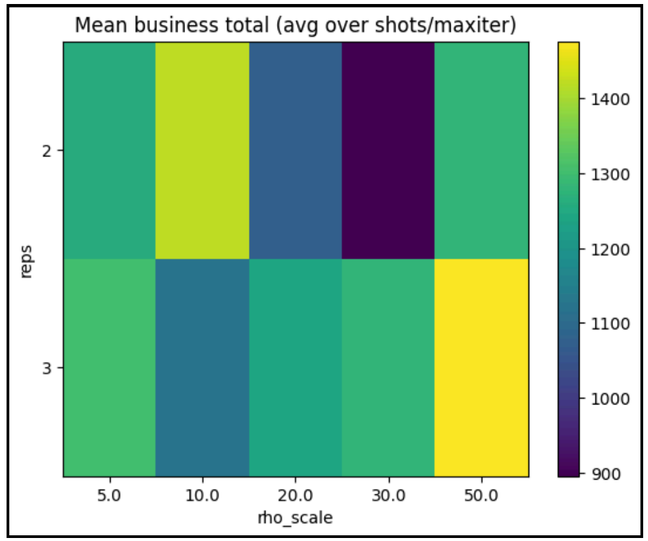
This is a Trotterized version of adiabatic evolution.

QAOA and its great use of physics



Results





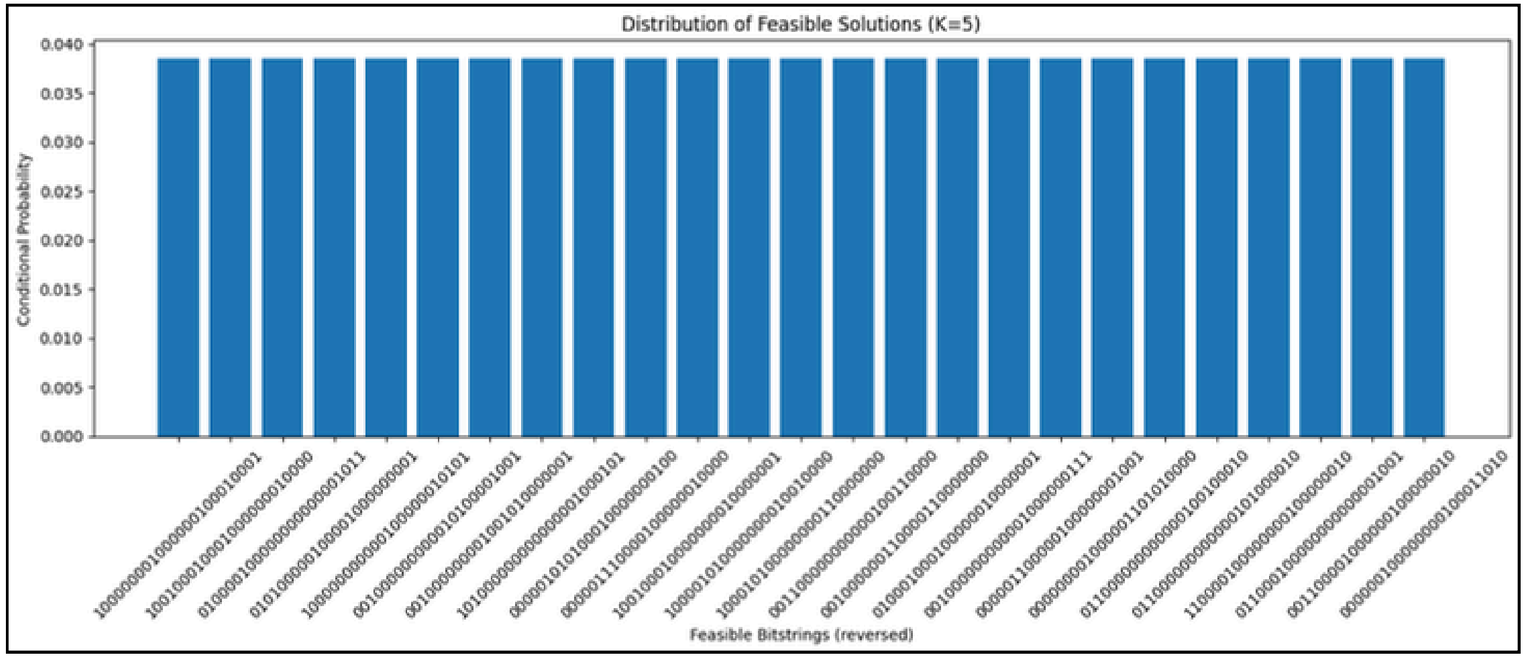
$$H_M^{(X)} = \sum_{i=1}^{n} X_i$$

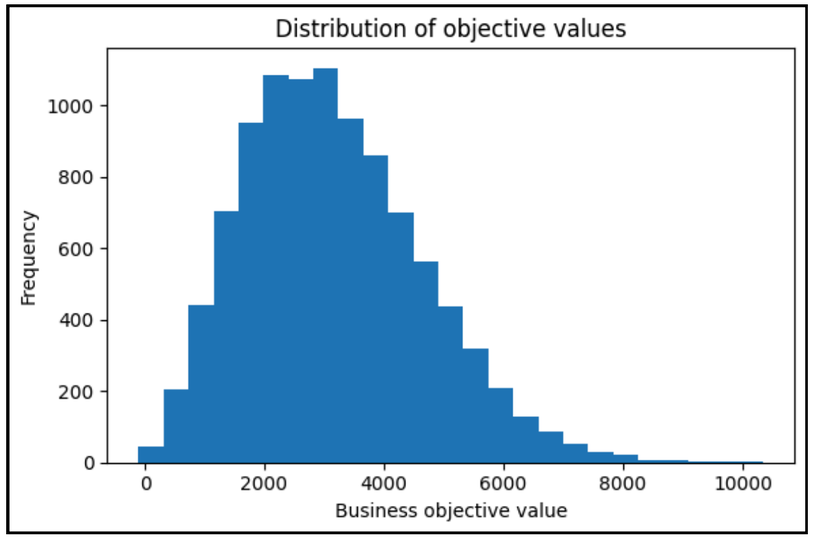
X-mixer

Results

Ground truth obtained using brut force: [0, 2, 3, 8, 10]

Results obtained: [0, 1, 6, 10, 22]





QAOA: tuned MIXER

Formulate classical optimization problem

1. Convert to QUBO
2. Map to Ising Hamiltonian
3. Build constrained QAOA circuit:
   - Feasible initial state
   - Cost Hamiltonian
   - Structured XY mixer
4. Optimize parameters classically
5. Sample final quantum distribution
6. Extract best feasible solution

$$e^{-i\beta(X_iX_j+Y_iY_j)}$$

Implements a partial SWAP between qubits i and j.

Correlation-based graph: Build edges using strong correlations in W

$$E = \text{ring} \cup \text{strong correlations from } W$$

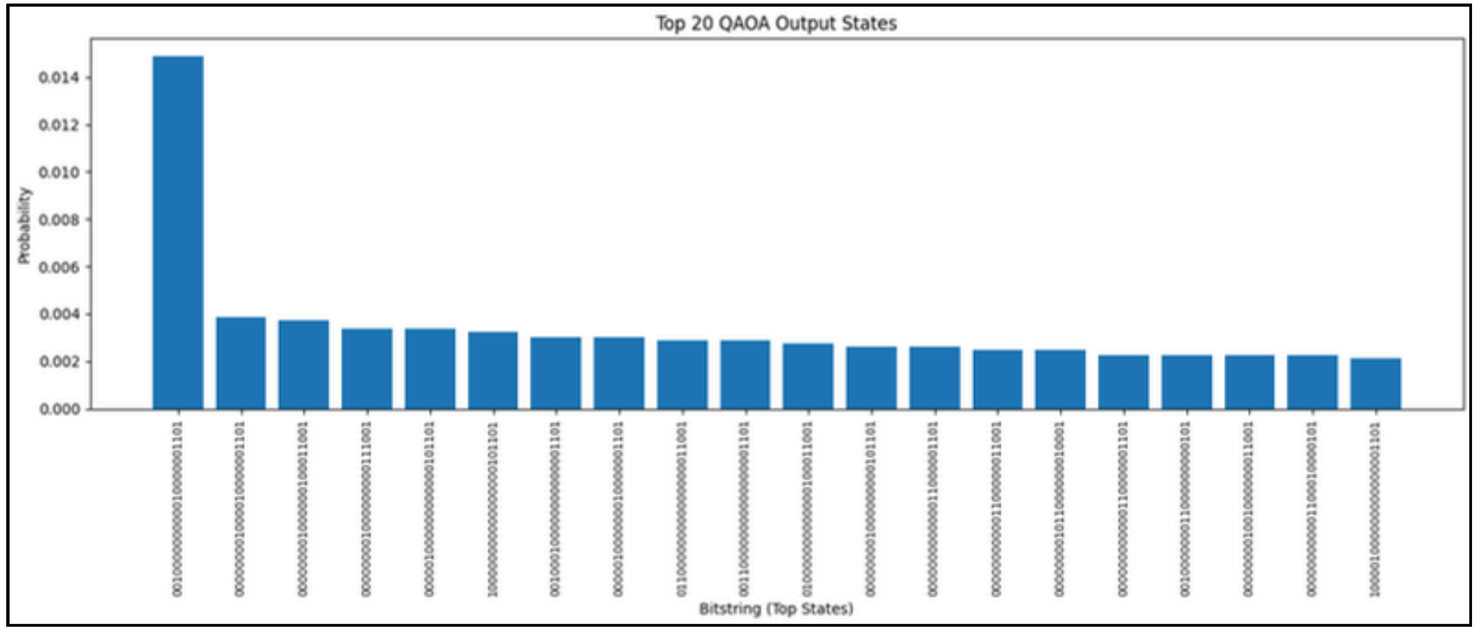
Tokens can easily move between variables that strongly interact in the objective

Creates superpositions only among states that satisfy the sparsity constraint

Results

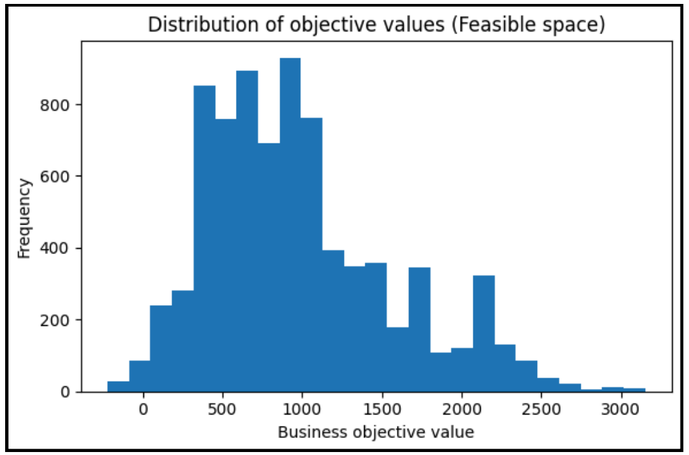
Ground truth obtained using brut force: [0, 2, 3, 8, 10]

Results obtained: Brut force: [0, 2, 3, 8, 10]



$$H_M^{(XY)} = \sum_{(i,j)\in E} (X_i X_j + Y_i Y_j)$$

XY-mixer



Conclusion

Quantum optimization represents exponentially many solutions within a single quantum state
It exploits interference rather than sequential evaluation
Constraints can be enforced structurally through symmetry preserving mixers

Quantum advantage does not come from running QAOA alone
It requires careful design of:
- The problem encoding
- The constraint handling
- The mixer structure

Constrained, problem-aware quantum circuits are key to **scalable** and efficient hybrid quantum optimization